

02_knn

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1 MA3622: Practical Machine Learning

variationalform https://variationalform.github.io/courses_PML/

Just Enough: progress at pace <https://github.com/variationalform>

Simon Shaw - <https://www.brunel.ac.uk/people/simon-shaw>.

2 k -NN's: k -Nearest Neighbours

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This document uses python

and also makes use of LaTeX

in Markdown

Google and possibly other search engines, both with and without AI enhancement, may have been used in the production of these notes.

AI tools such as Gemini, Claude, ChaptGPT etc may also have played a role in generating ideas and raw material for the following content.

Specific inclusions and sources are always referenced.

However, all planning and quality control is purely human, and my responsibility.

2.1 What this is about:

You will be introduced to ...

- The penguins data set, data frames, data selection
- Data engineering: *mean imputation*, and dropping unknowns
- Data bifurcation and trifurcation; calibration; tuning and **hyperparameters**
- k -Nearest Neighbours - classifying by *nearness*
- using the `KNeighborsClassifier` from `sklearn.neighbors`

- Confusion Matrices

The idea is that by using vectors to represent our data set, we can classify a new data point by finding the nearest data point to it for which the class is known. We then assign the new point with the same class.

Our emphasis is on *doing* rather than *proving* - **just enough: progress at pace.**

2.2 Assigned Reading

For further information and background refer to pages 19 - 25 of

- MLF: Machine Learning: A First Course for Engineers and Scientists, by Andreas Lindholm, Niklas Wahlström, Fredrik Lindsten, Thomas B. Schön. Cambridge University Press. <http://smlbook.org>.

The pages leading up to Page 19 are also highly recommended as an overview of concepts, purpose and uses of Machine Learning.

3 Penguins: An Example Data Set

We bring in our standard imports and then recall the data sets that are available in seaborn. We'll be using the *penguins* data.

```
[1]: import matplotlib.pyplot as plt
import numpy as np
from sklearn import datasets, linear_model
import pandas as pd
import seaborn as sns
```

```
[2]: # See, for example,
#     https://github.com/mwaskom/seaborn-data
#     https://blog.enterprisedna.co/how-to-load-sample-datasets-in-python/
datasets = sns.get_dataset_names()
for dataset in datasets:
    print(dataset, end=', ')
```

anagrams, anscombe, attention, brain_networks, car_crashes, diamonds, dots, dowjones, exercise, flights, fmri, geyser, glue, healthexp, iris, mpg, penguins, planets, seaice, taxis, tips, titanic,

3.1 Some Data-Engineering

As we have seen, there are a lot of data sets here that can be used to demonstrate various aspects of, and techniques in, Machine Learning and Data Science, and we'll look at a few of them - and others - as we progress.

To start with though we'll be working with the penguins data set. Before we do any machine learning we are going to have to do some **data cleaning**, see e.g. https://en.wikipedia.org/wiki/Data_cleansing, to remove some undefined values.

This shouldn't be confused with https://en.wikipedia.org/wiki/Feature_engineering.

Let's grab the penguins data and see what is in it. We load it into a data frame called `dfp`, as in *data frame for penguins*, and then look at the head of the table - the first few rows.

```
[3]: dfp = sns.load_dataset('penguins')
      dfp.head()
```

```
[3]:  species      island  bill_length_mm  bill_depth_mm  flipper_length_mm  \
0  Adelie  Torgersen         39.1           18.7           181.0
1  Adelie  Torgersen         39.5           17.4           186.0
2  Adelie  Torgersen         40.3           18.0           195.0
3  Adelie  Torgersen          NaN           NaN           NaN
4  Adelie  Torgersen         36.7           19.3           193.0

      body_mass_g      sex
0         3750.0    Male
1         3800.0  Female
2         3250.0  Female
3            NaN     NaN
4         3450.0  Female
```

Let's look at the shape of the data set - how many rows and columns does it have?

```
[4]: num_rows, num_columns = dfp.shape
      print('number of data points (or observations) = ', num_rows)
      print('number of features (or measurements) = ', num_columns)
```

```
number of data points (or observations) = 344
number of features (or measurements) = 7
```

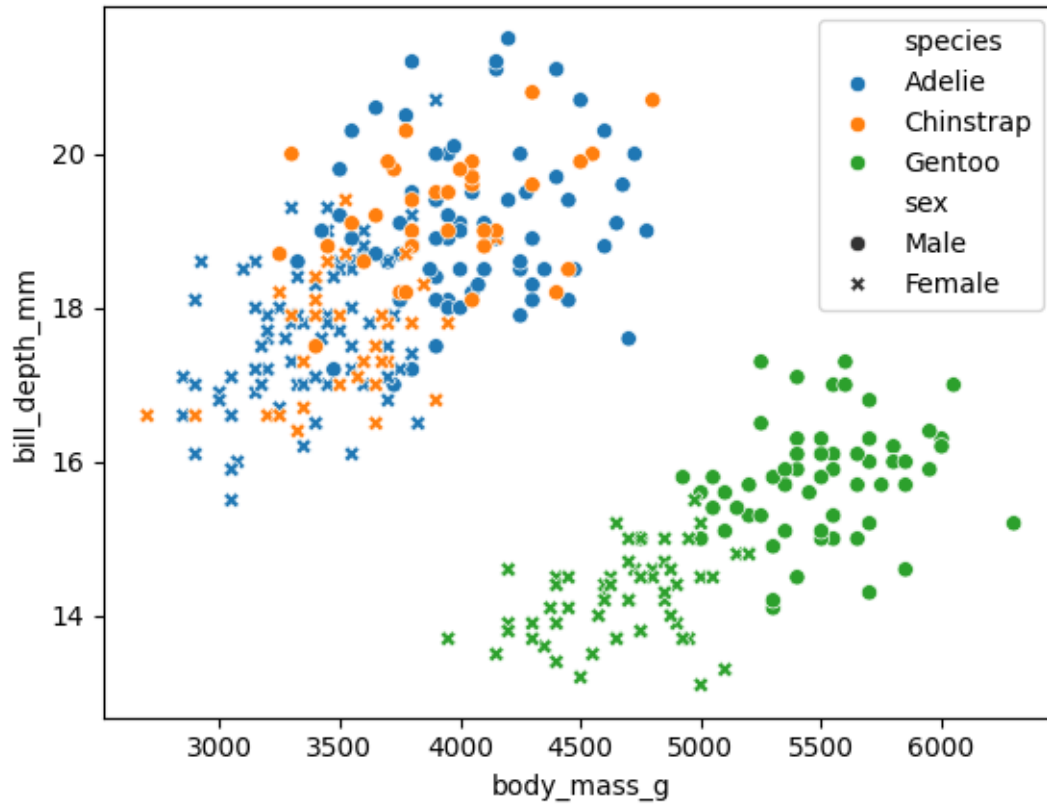
So, the data set contains 344 rows and seven columns. Each row corresponds to a single penguin, and for each row each column corresponds to a feature of that penguin. We can see its species, the island it was found on, its bill length, bill depth, and flipper length - all in millimetres, its body mass in grams, and its gender.

We can also see NaN values in row 3. That's the fourth row - be careful of this, indexing starts at zero. This stands for *Not a Number* and means that we can't use those values as they stand. We don't know why they are there - perhaps the data got corrupted. It's a fact of life though that data sets are often a bit messy with wrong, missing or corrupted values. We'll see a couple of ways to deal with these instances below.

We haven't listed every row - just the `head` of the data table. Another way to visualize these data is to use a scatter plot.

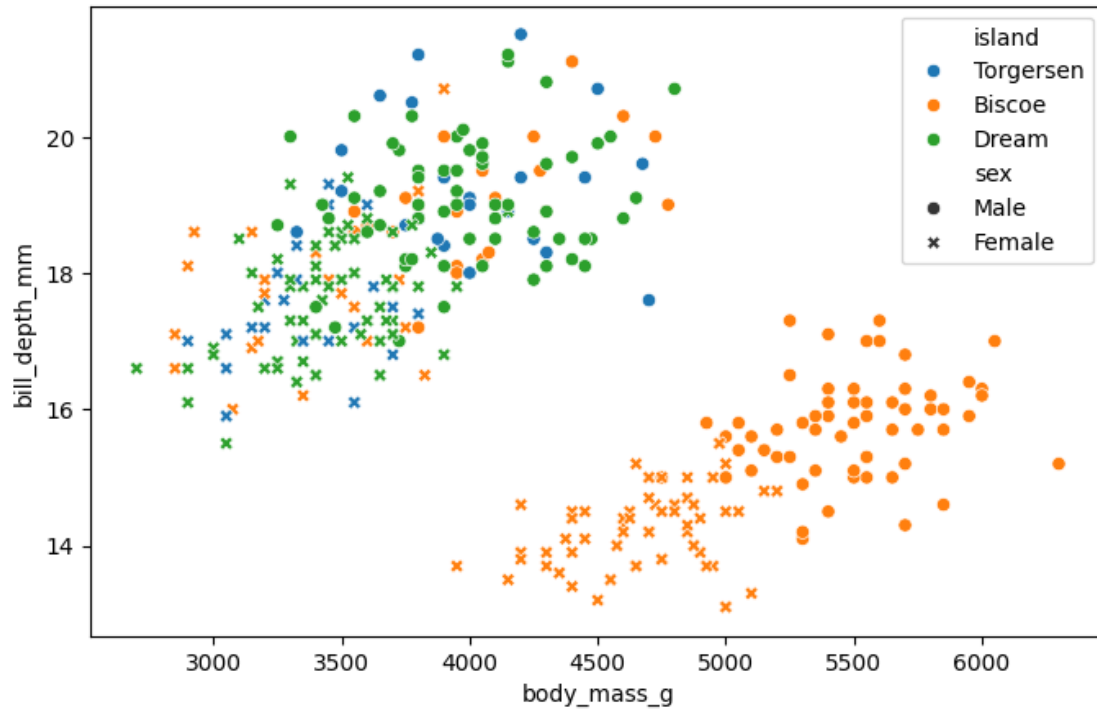
See e.g. <https://seaborn.pydata.org/generated/seaborn.scatterplot.html>

```
[5]: sns.scatterplot(data=dfp, x="body_mass_g", y="bill_depth_mm",
                    hue="species", style="sex");
```



If that looks a little cramped you can control the size like this:

```
[6]: plt.figure(figsize=(8, 5))
sns.scatterplot(data=dfp, x="body_mass_g", y="bill_depth_mm",
                hue="island", style="sex");
```



When we issued the command `dfp.head()` above we got to see the top of the table. We can also see the bottom like this:

```
[7]: dfp.tail()
```

```
[7]:
```

	species	island	bill_length_mm	bill_depth_mm	flipper_length_mm	\
339	Gentoo	Biscoe	NaN	NaN	NaN	
340	Gentoo	Biscoe	46.8	14.3	215.0	
341	Gentoo	Biscoe	50.4	15.7	222.0	
342	Gentoo	Biscoe	45.2	14.8	212.0	
343	Gentoo	Biscoe	49.9	16.1	213.0	

	body_mass_g	sex
339	NaN	NaN
340	4850.0	Female
341	5750.0	Male
342	5200.0	Female
343	5400.0	Male

- This has given us two species, *Adelie* and *Gentoo*, but from the plots above we know there is also a third: *Chinstrap*.
- We can also see from the *head* and *tail* functions that there are two islands, *Torgersen* and *Biscoe*, and that - from the plots - there is a third, *Dream*.
- How could we find these without having to plot the data?

A simple way is to ask for all the unique entries in the *species* column, and in the *island* column:

```
[8]: dfp.species.unique()
```

```
[8]: array(['Adelie', 'Chinstrap', 'Gentoo'], dtype=object)
```

```
[9]: dfp.island.unique()
```

```
[9]: array(['Torgersen', 'Biscoe', 'Dream'], dtype=object)
```

3.2 Summary

We have seen that three species are documented on three Antarctic islands.

We have also seen that some values are undefined: NaN stands for *Not a Number*. This may indicate that the data was not captured reliably.

We can see how many rows contain undefined values with this command:

```
[10]: dfp.isna().sum()
```

```
[10]: species          0
      island          0
      bill_length_mm    2
      bill_depth_mm     2
      flipper_length_mm  2
      body_mass_g       2
      sex              11
      dtype: int64
```

There are at least eleven - and there could be 11+2+2+2+2. Let's find them.

In the following `axis=1` tells python that we want to find **rows** with NaN in, as opposed to **columns**.

```
[11]: dfp[dfp.isna().any(axis=1)]
```

```
[11]:
```

	species	island	bill_length_mm	bill_depth_mm	flipper_length_mm	\
3	Adelie	Torgersen	NaN	NaN	NaN	
8	Adelie	Torgersen	34.1	18.1	193.0	
9	Adelie	Torgersen	42.0	20.2	190.0	
10	Adelie	Torgersen	37.8	17.1	186.0	
11	Adelie	Torgersen	37.8	17.3	180.0	
47	Adelie	Dream	37.5	18.9	179.0	
246	Gentoo	Biscoe	44.5	14.3	216.0	
286	Gentoo	Biscoe	46.2	14.4	214.0	
324	Gentoo	Biscoe	47.3	13.8	216.0	
336	Gentoo	Biscoe	44.5	15.7	217.0	
339	Gentoo	Biscoe	NaN	NaN	NaN	

	body_mass_g	sex
3	NaN	NaN
8	3475.0	NaN
9	4250.0	NaN
10	3300.0	NaN
11	3700.0	NaN
47	2975.0	NaN
246	4100.0	NaN
286	4650.0	NaN
324	4725.0	NaN
336	4875.0	NaN
339	NaN	NaN

We can get a list of the row index numbers like this:

```
[12]: NaN_rows = dfp[dfp.isna().any(axis=1)]
      print(NaN_rows.index)
```

```
Index([3, 8, 9, 10, 11, 47, 246, 286, 324, 336, 339], dtype='int64')
```

And we can use these as an alternative to the `axis=1` command above by typing

```
dfp.loc[NaN_rows.index]
```

```
[13]: dfp.loc[NaN_rows.index]
```

```
[13]: species      island  bill_length_mm  bill_depth_mm  flipper_length_mm  \
3      Adelie  Torgersen             NaN             NaN             NaN
8      Adelie  Torgersen             34.1             18.1             193.0
9      Adelie  Torgersen             42.0             20.2             190.0
10     Adelie  Torgersen             37.8             17.1             186.0
11     Adelie  Torgersen             37.8             17.3             180.0
47     Adelie    Dream             37.5             18.9             179.0
246   Gentoo   Biscoe             44.5             14.3             216.0
286   Gentoo   Biscoe             46.2             14.4             214.0
324   Gentoo   Biscoe             47.3             13.8             216.0
336   Gentoo   Biscoe             44.5             15.7             217.0
339   Gentoo   Biscoe             NaN             NaN             NaN
```

	body_mass_g	sex
3	NaN	NaN
8	3475.0	NaN
9	4250.0	NaN
10	3300.0	NaN
11	3700.0	NaN
47	2975.0	NaN
246	4100.0	NaN
286	4650.0	NaN
324	4725.0	NaN

```

336      4875.0  NaN
339      NaN   NaN

```

3.2.1 Data Engineering - our first method

One way to deal with missing values like this is to simply fill them with ‘reasonable’ values. For example, we can replace the numerical values with the mean, or average, of that feature, and replace categorical values with just one of the possible categories.

For example, let’s use the mean for numerical values and treat all missing genders as *Female*.

```

[14]: # from https://datagy.io/pandas-fillna/
dfp1 = dfp.fillna({'bill_length_mm' : dfp['bill_length_mm'].mean(),
                  'bill_depth_mm'   : dfp['bill_depth_mm'].mean(),
                  'flipper_length_mm': dfp['flipper_length_mm'].mean(),
                  'body_mass_g'      : dfp['body_mass_g'].mean(),
                  'sex': 'Female'})

```

Mean Imputation Replacing a missing numerical feature value with the mean of the known feature values in this way is called **imputing the mean**. It is easy to implement - just one line above - but you should be aware that it corrupts the original data set.

- On the upside this process maintains the sample size
- On the downside it (probably) alters some statistical properties of the data (the unknown variance, for example).

As an analyst you would be responsible for taking a decision as to how to deal with missing values. You may not be the only one involved in that decision.

We can compare the old and new data frames just to check this worked as expected.

```

[15]: # Here is the new one with the NaN's replaced
dfp1.loc[NaN_rows.index]

```

```

[15]:   species  island  bill_length_mm  bill_depth_mm  flipper_length_mm  \
3   Adelie  Torgersen      43.92193      17.15117      200.915205
8   Adelie  Torgersen      34.10000      18.10000      193.000000
9   Adelie  Torgersen      42.00000      20.20000      190.000000
10  Adelie  Torgersen      37.80000      17.10000      186.000000
11  Adelie  Torgersen      37.80000      17.30000      180.000000
47  Adelie   Dream      37.50000      18.90000      179.000000
246  Gentoo  Biscoe      44.50000      14.30000      216.000000
286  Gentoo  Biscoe      46.20000      14.40000      214.000000
324  Gentoo  Biscoe      47.30000      13.80000      216.000000
336  Gentoo  Biscoe      44.50000      15.70000      217.000000
339  Gentoo  Biscoe      43.92193      17.15117      200.915205

      body_mass_g  sex
3   4201.754386  Female

```

```

8    3475.000000  Female
9    4250.000000  Female
10   3300.000000  Female
11   3700.000000  Female
47   2975.000000  Female
246  4100.000000  Female
286  4650.000000  Female
324  4725.000000  Female
336  4875.000000  Female
339  4201.754386  Female

```

```

[16]: # Here is the old one with the NaN's
      dfp.loc[NaN_rows.index]

```

```

[16]:      species      island  bill_length_mm  bill_depth_mm  flipper_length_mm  \
3    Adelie  Torgersen         NaN             NaN             NaN
8    Adelie  Torgersen        34.1             18.1             193.0
9    Adelie  Torgersen        42.0             20.2             190.0
10   Adelie  Torgersen        37.8             17.1             186.0
11   Adelie  Torgersen        37.8             17.3             180.0
47   Adelie    Dream        37.5             18.9             179.0
246  Gentoo   Biscoe        44.5             14.3             216.0
286  Gentoo   Biscoe        46.2             14.4             214.0
324  Gentoo   Biscoe        47.3             13.8             216.0
336  Gentoo   Biscoe        44.5             15.7             217.0
339  Gentoo   Biscoe         NaN             NaN             NaN

      body_mass_g  sex
3             NaN  NaN
8          3475.0  NaN
9          4250.0  NaN
10         3300.0  NaN
11         3700.0  NaN
47         2975.0  NaN
246         4100.0  NaN
286         4650.0  NaN
324         4725.0  NaN
336         4875.0  NaN
339           NaN  NaN

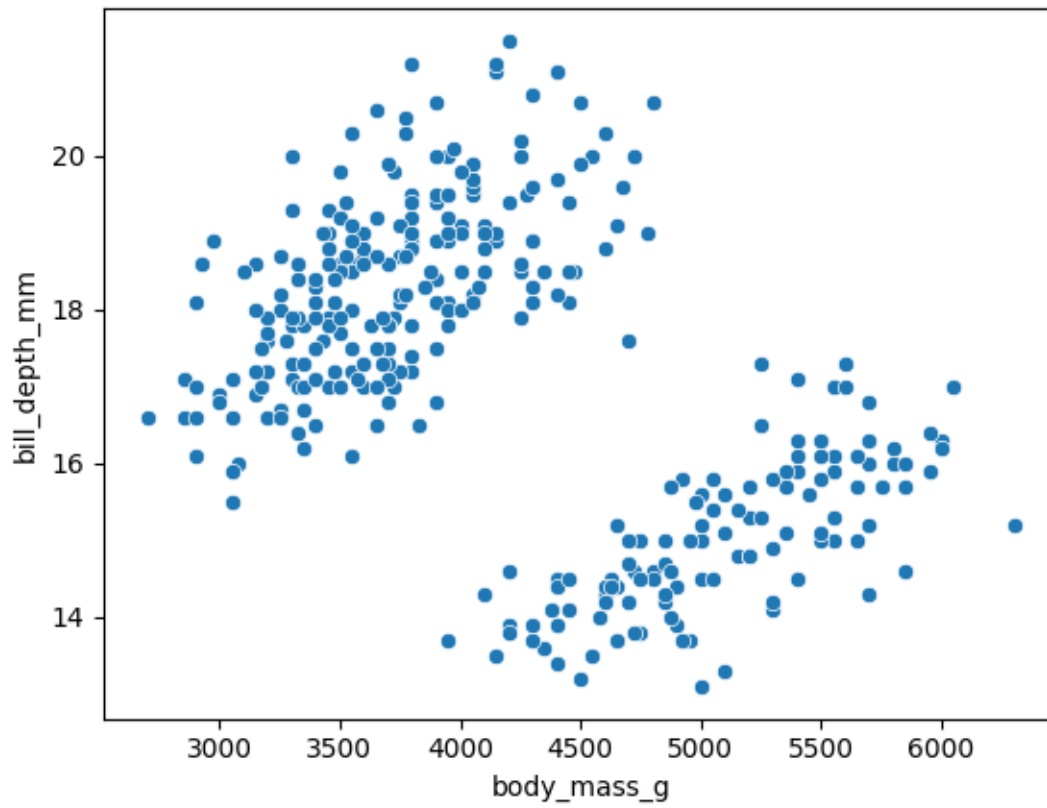
```

It is always good practice to check your work. This can be challenging when dealing with large data sets because you can't keep printing them out and checking every item to make sure that no errors have been introduced.

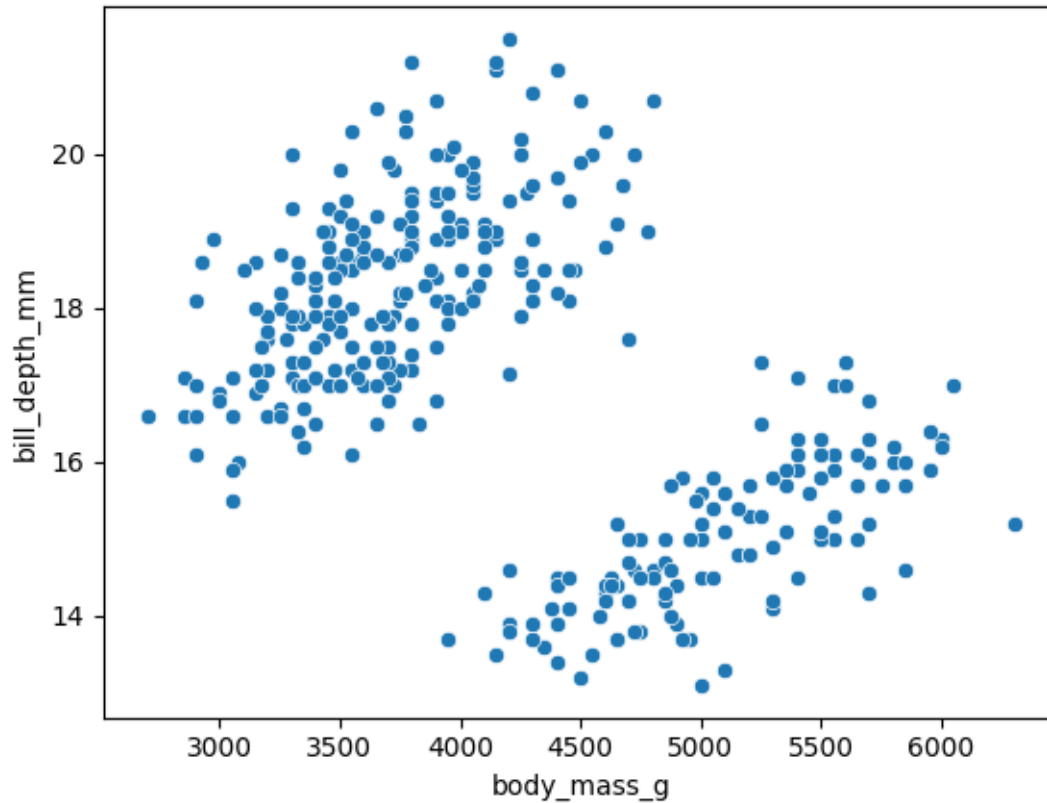
One way to make sure that these commands didn't do something unexpected behind the scenes is just to plot each data set and make sure they look the same.

For example:

```
[17]: sns.scatterplot(data=dfp, x="body_mass_g", y="bill_depth_mm");
```



```
[18]: sns.scatterplot(data=dfp1, x="body_mass_g", y="bill_depth_mm");
```



Alternatively, the `describe()` function prints summary statistics. These should be similar (probably not exactly the same) for each.

Below we see how this works. What do you think? Is everything broadly OK with our data set?

Can you explain the differences?

```
[19]: dfp.describe()
```

```
[19]:
```

	bill_length_mm	bill_depth_mm	flipper_length_mm	body_mass_g
count	342.000000	342.000000	342.000000	342.000000
mean	43.921930	17.151170	200.915205	4201.754386
std	5.459584	1.974793	14.061714	801.954536
min	32.100000	13.100000	172.000000	2700.000000
25%	39.225000	15.600000	190.000000	3550.000000
50%	44.450000	17.300000	197.000000	4050.000000
75%	48.500000	18.700000	213.000000	4750.000000
max	59.600000	21.500000	231.000000	6300.000000

```
[20]: dfp1.describe()
```

```
[20]:      bill_length_mm  bill_depth_mm  flipper_length_mm  body_mass_g
count      344.000000      344.000000      344.000000      344.000000
mean       43.921930      17.151170      200.915205     4201.754386
std        5.443643       1.969027       14.020657       799.613058
min        32.100000      13.100000      172.000000     2700.000000
25%        39.275000      15.600000      190.000000     3550.000000
50%        44.250000      17.300000      197.000000     4050.000000
75%        48.500000      18.700000      213.000000     4750.000000
max        59.600000      21.500000      231.000000     6300.000000
```

3.2.2 Data Engineering - our second method

In the method above we just replaced missing values with (hopefully) *nearby* ones.

On the other hand, if we have a lot of data and are able to live with a little less of it then we can just drop the data items (rows) that contain one or more undefined values.

THINK ABOUT: what could go wrong?

For example: let's recall the rows with NaN entries and then total up how many there are in each column, and in total:

```
[21]: dfp.loc[NaN_rows.index]
```

```
[21]:      species      island  bill_length_mm  bill_depth_mm  flipper_length_mm  \
3      Adelie  Torgersen          NaN          NaN          NaN
8      Adelie  Torgersen         34.1          18.1          193.0
9      Adelie  Torgersen         42.0          20.2          190.0
10     Adelie  Torgersen         37.8          17.1          186.0
11     Adelie  Torgersen         37.8          17.3          180.0
47     Adelie      Dream         37.5          18.9          179.0
246    Gentoo    Biscoe         44.5          14.3          216.0
286    Gentoo    Biscoe         46.2          14.4          214.0
324    Gentoo    Biscoe         47.3          13.8          216.0
336    Gentoo    Biscoe         44.5          15.7          217.0
339    Gentoo    Biscoe          NaN          NaN          NaN

      body_mass_g  sex
3              NaN  NaN
8             3475.0  NaN
9             4250.0  NaN
10            3300.0  NaN
11            3700.0  NaN
47            2975.0  NaN
246            4100.0  NaN
286            4650.0  NaN
324            4725.0  NaN
336            4875.0  NaN
339              NaN  NaN
```

```
[22]: dfp.isna().sum()
```

```
[22]: species          0
      island          0
      bill_length_mm    2
      bill_depth_mm    2
      flipper_length_mm  2
      body_mass_g       2
      sex             11
      dtype: int64
```

We could have written `dfp.isna().sum(axis=0)` to insist that we are counting down columns here, but that's the default so the `axis=0` isn't needed.

We can see that there are no more than two NaN values in the third to sixth columns, but eleven in the last, the seventh, column.

NOTE: the digit in the left most column is just the index of the column - it is not considered part of the data set.

So, given that we have 344 data points (penguins), it looks like we can afford to drop these bad data rows from the set. We can do it like this:

```
[23]: dfp2 = dfp.dropna()
```

Let's compare...

```
[24]: dfp
```

```
[24]:
```

	species	island	bill_length_mm	bill_depth_mm	flipper_length_mm	\
0	Adelie	Torgersen	39.1	18.7	181.0	
1	Adelie	Torgersen	39.5	17.4	186.0	
2	Adelie	Torgersen	40.3	18.0	195.0	
3	Adelie	Torgersen	NaN	NaN	NaN	
4	Adelie	Torgersen	36.7	19.3	193.0	
..	...	...	...	...	...	
339	Gentoo	Biscoe	NaN	NaN	NaN	
340	Gentoo	Biscoe	46.8	14.3	215.0	
341	Gentoo	Biscoe	50.4	15.7	222.0	
342	Gentoo	Biscoe	45.2	14.8	212.0	
343	Gentoo	Biscoe	49.9	16.1	213.0	

	body_mass_g	sex
0	3750.0	Male
1	3800.0	Female
2	3250.0	Female
3	NaN	NaN
4	3450.0	Female
..	...	...

```

339      NaN      NaN
340    4850.0  Female
341    5750.0    Male
342    5200.0  Female
343    5400.0    Male

```

```
[344 rows x 7 columns]
```

```
[25]: dfp2
```

```

[25]:   species   island  bill_length_mm  bill_depth_mm  flipper_length_mm  \
0   Adelie  Torgersen         39.1           18.7             181.0
1   Adelie  Torgersen         39.5           17.4             186.0
2   Adelie  Torgersen         40.3           18.0             195.0
4   Adelie  Torgersen         36.7           19.3             193.0
5   Adelie  Torgersen         39.3           20.6             190.0
..    ...      ...
338  Gentoo   Biscoe         47.2           13.7             214.0
340  Gentoo   Biscoe         46.8           14.3             215.0
341  Gentoo   Biscoe         50.4           15.7             222.0
342  Gentoo   Biscoe         45.2           14.8             212.0
343  Gentoo   Biscoe         49.9           16.1             213.0

```

```

      body_mass_g    sex
0      3750.0    Male
1      3800.0  Female
2      3250.0  Female
4      3450.0  Female
5      3650.0    Male
..      ...      ...
338     4925.0  Female
340     4850.0  Female
341     5750.0    Male
342     5200.0  Female
343     5400.0    Male

```

```
[333 rows x 7 columns]
```

It looks fine - the NaN values have disappeared from the newly engineered dataset. We can check, as above, by counting how many NaN's are found in the new data set:

```
[26]: dfp2.isna().sum()
```

```

[26]: species      0
      island      0
      bill_length_mm  0
      bill_depth_mm  0

```

```
flipper_length_mm    0
body_mass_g          0
sex                  0
dtype: int64
```

On the other hand, the index values in the left most column are off. There is no **3** for example. We can reset them with the `reset_index()` function but we have to make sure we drop the original indices otherwise they will persist.

```
[27]: dfp2 = dfp2.reset_index(drop=True); dfp2
```

```
[27]:
```

	species	island	bill_length_mm	bill_depth_mm	flipper_length_mm	\
0	Adelie	Torgersen	39.1	18.7	181.0	
1	Adelie	Torgersen	39.5	17.4	186.0	
2	Adelie	Torgersen	40.3	18.0	195.0	
3	Adelie	Torgersen	36.7	19.3	193.0	
4	Adelie	Torgersen	39.3	20.6	190.0	
..	...	...	...	...	...	
328	Gentoo	Biscoe	47.2	13.7	214.0	
329	Gentoo	Biscoe	46.8	14.3	215.0	
330	Gentoo	Biscoe	50.4	15.7	222.0	
331	Gentoo	Biscoe	45.2	14.8	212.0	
332	Gentoo	Biscoe	49.9	16.1	213.0	

	body_mass_g	sex
0	3750.0	Male
1	3800.0	Female
2	3250.0	Female
3	3450.0	Female
4	3650.0	Male
..	...	...
328	4925.0	Female
329	4850.0	Female
330	5750.0	Male
331	5200.0	Female
332	5400.0	Male

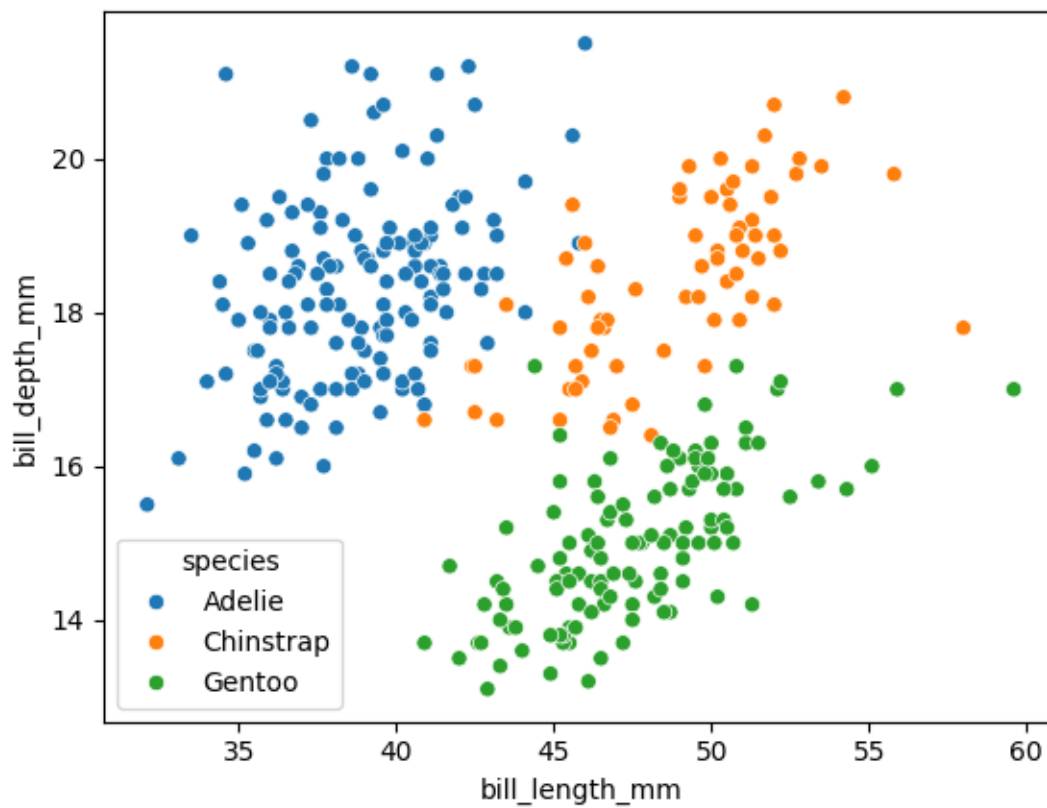
```
[333 rows x 7 columns]
```

Now we have a clean data set with no false values introduced, with no undefined entries, and with consecutive labelling down the left.

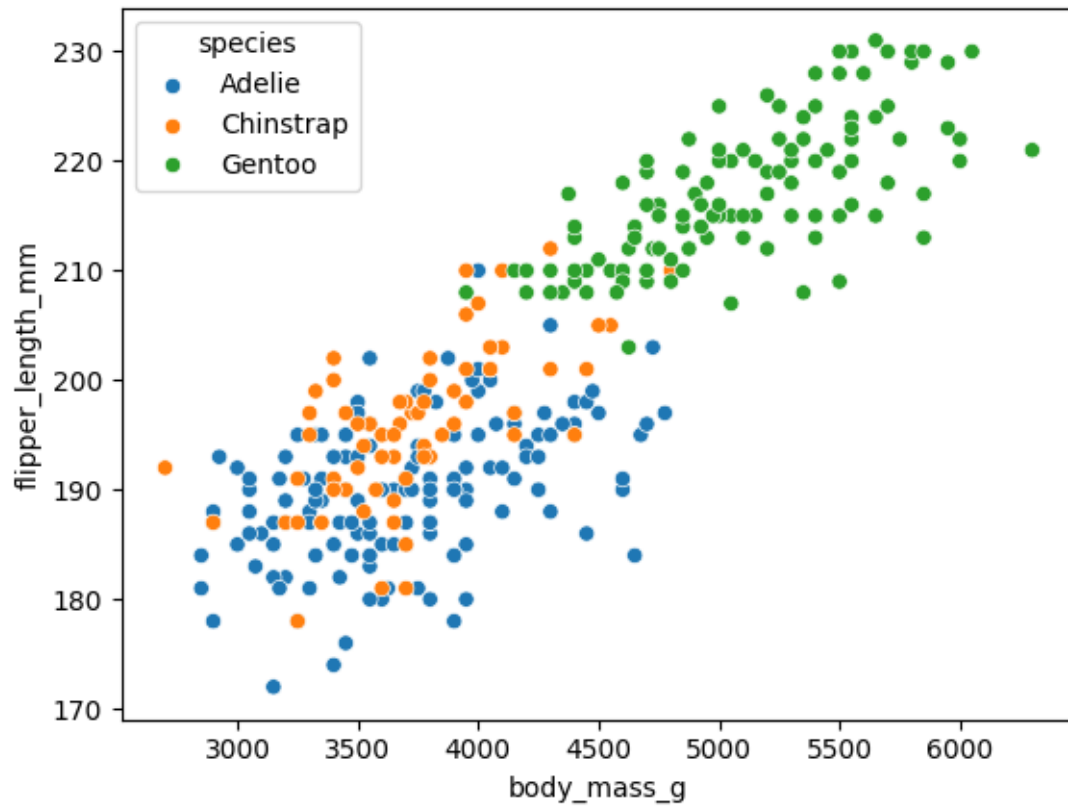
Visualization Data sets are often much too large to be able to effectively work with them in tabular form. Visualization is then more useful.

Let's pause to explore a few visuals of our cleaned-up data set.

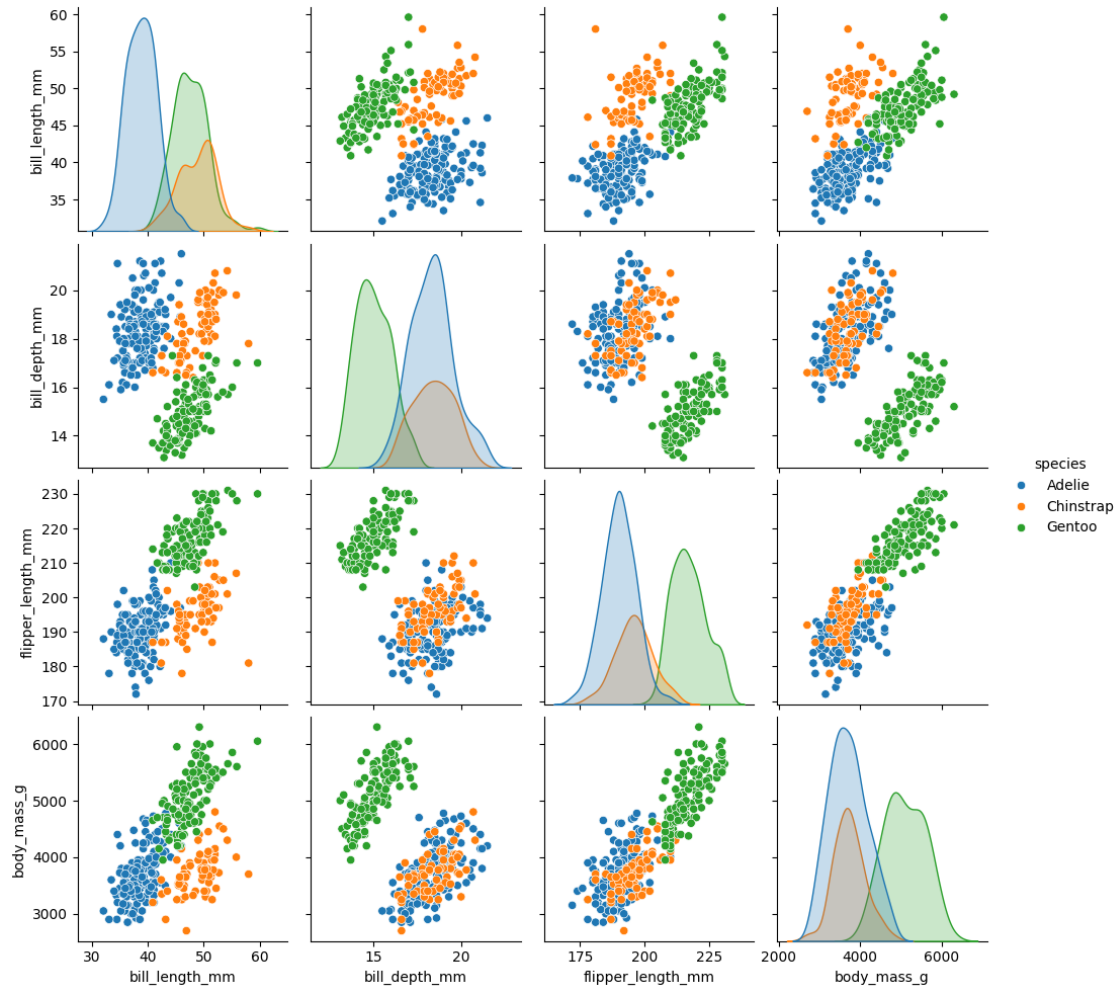
```
[28]: sns.scatterplot(data=dfp2, x="bill_length_mm", y="bill_depth_mm",  
                    hue="species");
```



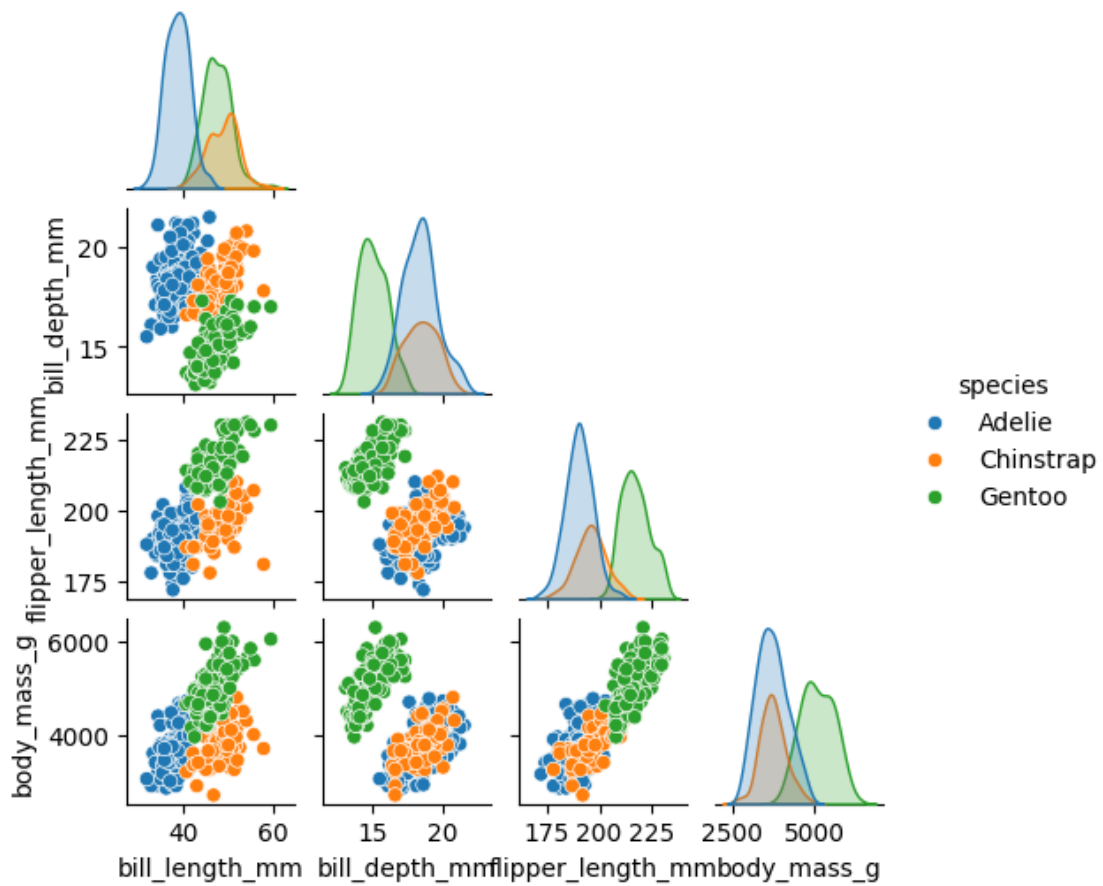
```
[29]: sns.scatterplot(data=dfp2, x="body_mass_g", y="flipper_length_mm",  
                    hue="species");
```



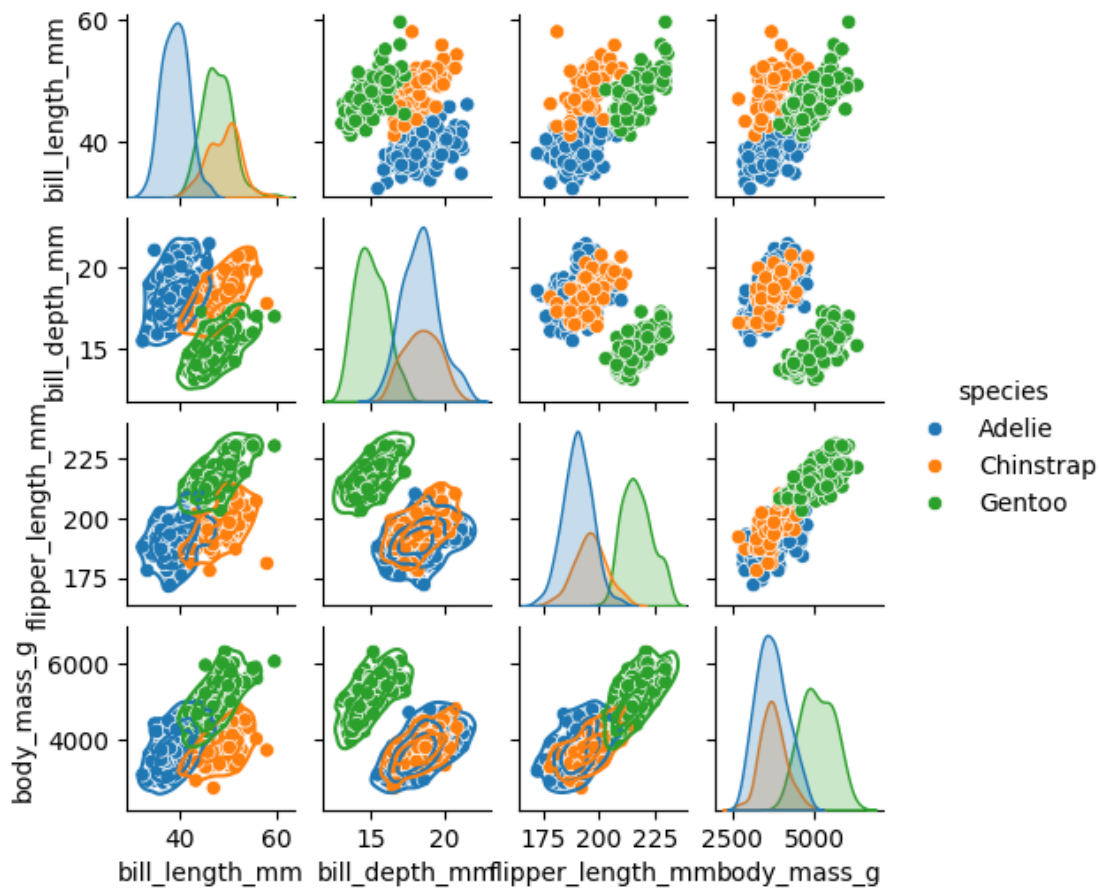
```
[30]: sns.pairplot(dfp2, hue='species');
```



```
[31]: # lots of options for the above. See
      # https://seaborn.pydata.org/generated/seaborn.pairplot.html
      sns.pairplot(dfp2, corner=True, hue='species', height=1.3);
```



```
[32]: g = sns.pairplot(dfp2, diag_kind="kde", hue='species', height=1.3)
      g.map_lower(sns.kdeplot, levels=4, color=".2");
```



Further Exploration of the Data Set So far we have loaded the data, and operated on it row by row as well as plotted various views of the data.

Let's look now at how to manipulate the data set at a lower level, and see how we might separate out clusters of data - data items that each share a common feature.

Recall, this is what our set contains...

```
[33]: dfp2.head()
```

```
[33]:  species      island  bill_length_mm  bill_depth_mm  flipper_length_mm  \
0  Adelie  Torgersen         39.1           18.7           181.0
1  Adelie  Torgersen         39.5           17.4           186.0
2  Adelie  Torgersen         40.3           18.0           195.0
3  Adelie  Torgersen         36.7           19.3           193.0
4  Adelie  Torgersen         39.3           20.6           190.0

      body_mass_g      sex
0         3750.0    Male
1         3800.0  Female
```

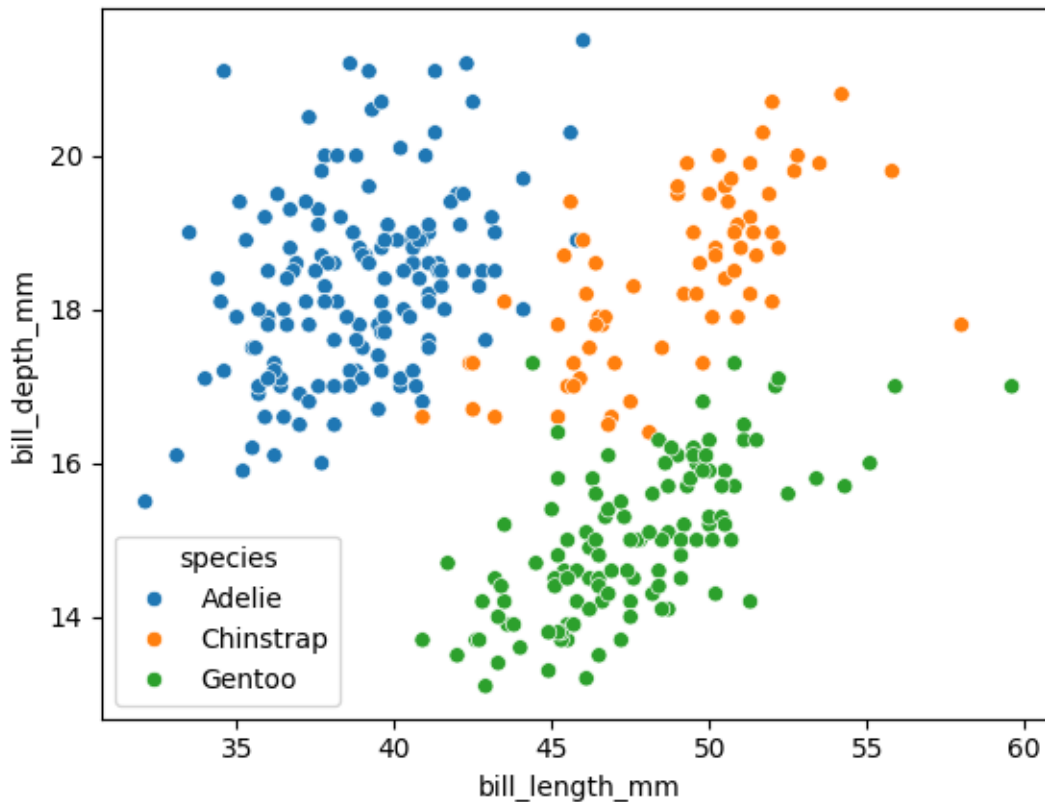
```

2      3250.0  Female
3      3450.0  Female
4      3650.0   Male

```

We can see how the species form almost distinct clusters with the following plot.

```
[34]: sns.scatterplot(data=dfp2, x="bill_length_mm", y="bill_depth_mm",
                      hue="species");
```



We can access the column of `species` data using square brackets like this

```
dfp2['species']
```

This refers to every row - with lots of repeated values.

We can squeeze out the repeats into just one uniquely occurring feature value like this...

```
[35]: dfp2['species'].unique()
```

```
[35]: array(['Adelie', 'Chinstrap', 'Gentoo'], dtype=object)
```

This tells us that there are three unique species. We knew this from the plots - but that was a human taking a look. This method allows the code to determine the same information without

human intervention.

Creating Data Subsets It is sometimes useful to be able to separate out the data subsets, by a given feature value. If we choose to separate by 'species' then this command

```
dfp2.loc[ dfp2['species'] == 'Adelie' ]
```

will give us back a new data frame that just contains the Adelie penguin data. It does this by using square brackets and double equals so that this statement,

```
dfp2['species'] == 'Adelie'
```

evaluates to **true** if, for a given row, the species feature is *Adelie*. Then

```
dfp2.loc[ ? ]
```

keeps only those rows for which the question mark is *true*. We can assign these rows to a new data frame.

This means that we can create three data subsets - one for each species - as follows...

```
[36]: dfA = dfp2.loc[dfp2['species'] == 'Adelie']
      dfC = dfp2.loc[dfp2['species'] == 'Chinstrap']
      dfG = dfp2.loc[dfp2['species'] == 'Gentoo']
```

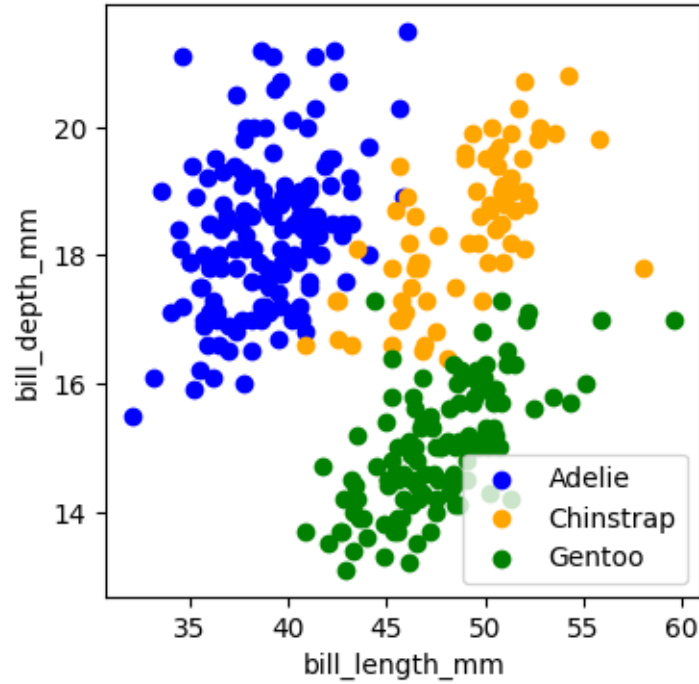
Using matplotlib to plot the clusters separately We can use `plt.scatter` to plot scatter plots directly in matplotlib as below. First we create arrays (vectors if you like) of values, and then we plot them in 2D.

```
[37]: blA=np.array(dfA['bill_length_mm'].tolist())
      bdA=np.array(dfA['bill_depth_mm'].tolist())

      blC=np.array(dfC['bill_length_mm'].tolist())
      bdC=np.array(dfC['bill_depth_mm'].tolist())

      blG=np.array(dfG['bill_length_mm'].tolist())
      bdG=np.array(dfG['bill_depth_mm'].tolist())
```

```
[38]: plt.rcParams["figure.figsize"] = (4,4)
      plt.scatter(blA,bdA,color='blue')
      plt.scatter(blC,bdC,color='orange')
      plt.scatter(blG,bdG,color='green')
      plt.xlabel('bill_length_mm')
      plt.ylabel('bill_depth_mm')
      plt.legend(['Adelie', 'Chinstrap', 'Gentoo'],loc='lower right');
```



3.3 k -NN's - developing intuition

We can now look at the k Nearest Neighbours, or k -NN, method for classification of data. The setting we assume at the outset is that we have a ‘training set’ of data such that each row of the data set corresponds to one observation.

Moreover, in each row there are numerical features which can be organized into a vector, $x = (x_1, x_2, \dots, x_n)^T$, and a label, y , which is categorical.

There may be other numerical and categorical data that we choose not to use.

We imagine plotting these data points in n -dimensional space (hard to imagine when $n > 3$, which is why the abstraction of mathematics is so useful), and we imagine them being coloured according to the value of the label y .

In the example above we had

$$x = (\text{bill_length_mm}, \text{bill_depth_mm})^T \quad (1)$$

$$y = (\text{Adelie}, \text{Chinstrap}, \text{Gentoo})^T \quad (2)$$

and we coloured the labels as blue, orange or green.

Now imagine that a field researcher reports in some new measurements for a penguin, and that we want to classify its species based only on those measurements.

The idea is to plot the new measurements and see which cluster of like colour they are closest to. This closest cluster (colour) is then used to assign the species to that new measurement.

Let's see a dummy run of this in a picture.

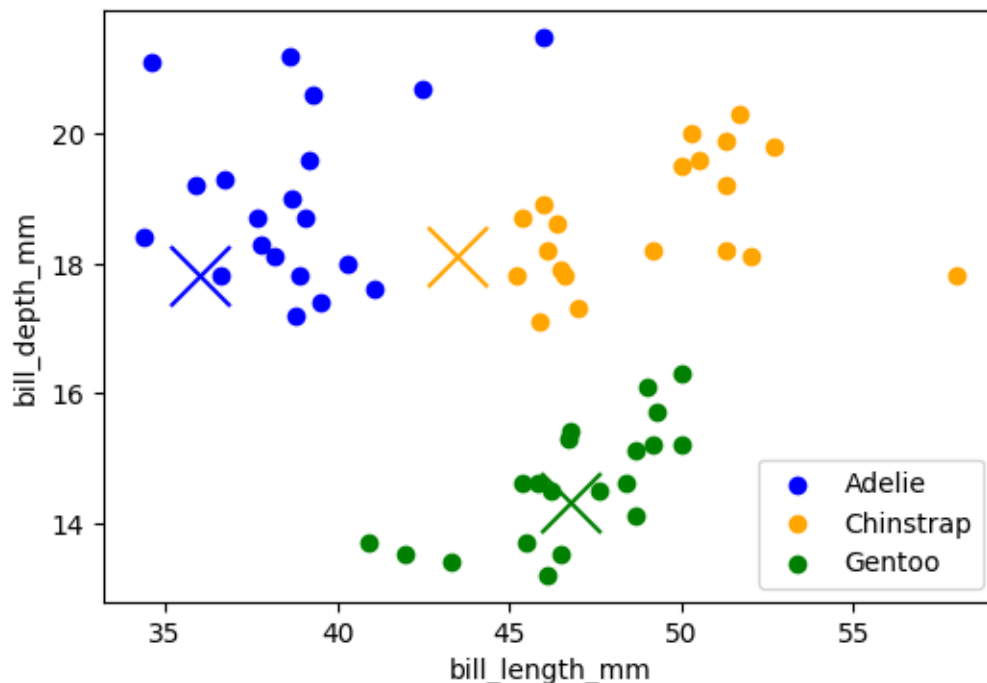
In the diagram below we pretend that we only have the first twenty rows of each of the data subsets. We plot them as coloured dots, just as above.

Then we pretend that we get three new observations. For illustration purposes we take the entries from the fourth from last position in each data set.

But in the **REAL WORLD** we would be expecting new data to be arriving **UNSEEN** from the field.

We plot these 'new observations' with a cross.

```
[39]: plt.figure(figsize=(6, 4))
# plot first twenty rows of each as coloured dots.
plt.scatter(blA[0:20],bdA[0:20],color='blue')
plt.scatter(blC[0:20],bdC[0:20],color='orange')
plt.scatter(blG[0:20],bdG[0:20],color='green')
plt.legend(['Adelie', 'Chinstrap', 'Gentoo'],loc='lower right')
plt.xlabel('bill_length_mm'); plt.ylabel('bill_depth_mm')
indx = -4 # plot data item fourth from the end in each as a cross
plt.scatter(blA[indx],bdA[indx],color='blue', marker='x', s=500)
plt.scatter(blC[indx],bdC[indx],color='orange', marker='x', s=500)
plt.scatter(blG[indx],bdG[indx],color='green', marker='x', s=500);
```



We carry out the classification as follows:

1. The green cross is quite central in the green, Gentoo, cluster and so we can classify this new observation as a Gentoo penguin.
2. The blue cross isn't that central in the blue cluster, but on the other hand it is far away from the yellow and green clusters and so we can safely classify this observation as an Adelie penguin.
3. The yellow cross presents us with more of a dilemma though. A careful look suggests that it is slightly closer to the yellow cluster than the blue and so, on that basis, we would probably choose to classify that penguin as a Chinstrap.

Any comments, thoughts, questions? The first two steps seem safe, and justifiable. They are *explainable*. The third less so. We can see that the yellow cross corresponds to a fairly typical bill depth for an Adelie.

- So is it a Chinstrap?
- We can also see that Adelie penguins have bill lengths that straddle the value indicated by the yellow cross.
- So should the yellow cross observation be classified as a Chinstrap?
- We see here that the issue of **explainability** can be vexed.
- If we had more data the yellow cross might become obviously a Chinstrap,
- Or it might be obvious that it is an Adelie.

Explainability may or may not matter. But it is increasingly becoming a hot topic in data science.

Suppose your pension fund invested everything in a new tech venture that was going to design batteries with infinite life. It will fail of course.

If this venture was suggested by an Artificially Intelligent agent powered by machine learning algorithms then the pension company directors won't be able to explain their reasoning if the underlying data science was not explainable.

This is hardly realistic, but explainability is a big and important deal in areas like finance and investing, and in medical diagnosis, to name but two. The reasons for its importance are obvious.

4 k -NN's - the mathematical details

We index each data point in the training set with a subscript. So we have the feature vectors $x_1, x_2, x_3, \dots$. Each of these has a label, $y_1, y_2, y_3, \dots$

These are the coloured dots above. The positions are the features. The colours are the labels.

We now get a new observation, x^* and we want to classify it - we want to apply a label to it using the data from the training set.

The mathematical version of the process we followed above was to determine the distance between x^* and each x_i using

$$\|x^* - x_i\|_2 \quad (\text{recall: the Euclidean, Pythagorean or } \ell_2 \text{ norm}).$$

We then to choose the value i such that this distance is a minimum. The label, y_i , corresponding to that particular i is then assigned to the new observation x^* .

4.1 Cross-Reference to the Assigned Reading

You were recommended to read pages 19 - 25 of

- MLF: Machine Learning: A First Course for Engineers and Scientists, by Andreas Lindholm, Niklas Wahlström, Fredrik Lindsten, Thomas B. Schön. Cambridge University Press. <http://smlbook.org>.

More details on this are given there, in particular:

- the use of k -NN for regression as well as classification.
- the use of more than one ‘nearest’ neighbour - see which cluster ‘wins’ a vote.
- notes on how to choose the number of neighbours, and ‘overfitting’.
- the importance of normalizing the inputs

Also of importance, but not mentioned in the book, is the choice of norm. We referred to the Euclidean or Pythagorean norm above, but we could just as easily have chosen any of the other vector p norms. For example,

$\|\cdot\|_1$ - Manhattan, ‘taxicab’, norm; $\|\cdot\|_\infty$ - ‘infinity’, ‘max’, norm.

4.1.1 Hyperparameters

In the discussion above we just touched upon the important issue of picking *hyperparameters*. These are values and choices that need to be specified to the algorithm, the code, prior to the machine learning phase.

In the above we mentioned that we need to choose:

- k - the number of nearest neighbours to search for.
- p - the choice of norm to use to measure distance, *nearness*.

These are *human* choices: the *hyperparameters* are not learned from the data, but need to be chosen upfront.

4.1.2 Data Set Bifurcation and Trifurcation

However, we don’t necessarily need to worry about making a wrong choice of hyperparameters that cannot subsequently be changed. In practice we would be prepared to *calibrate* the model by *tuning* its performance by turning the dials on the hyperparameter values.

Usually the dataset that we are working with will be either *bifurcated* into a *training* and a *test* set. Or will be *trifurcated* into a *training*, *validation* and a *test* set.

We’ll return to this as we go through, but briefly...

- The *training set*: used to initialise the machine learning model.
- The *validation set*: used to tune the hyperparameters.

- The *test set*: used as **unseen data** to derive final performance quality measurements after training and validation has been completed.

It is important to realise that the test set output should never be used to further tune and calibrate the model. It is a **hold out** set that simulates how the model will perform in the **real world** on unseen data.

The data set is treated in all of these cases as *ground truth* - it is believed to be true, although in practice some data points might contain errors, or be missing. And there is almost certainly going to be some noise on any numerical values recorded in the data.

We never ask where that truth actually came from though... This might be tricky. We need some form of **gold standard** method that always produces correct labels for the data. You can imagine the issues around that assumption...

There no hard and fast rules on the proportions to use to bifurcate or trifurcate the data set. We might bifurcate using 75%/25% for example, or trifurcate with 50%/25%/25%.

4.2 Introducing `scikit-learn`

Let's now see how to use `scikit-learn` to do *k*-NN classification with the penguins data that we cleaned and prepared.

The following code was adapted in its early stages from *Machine Learning with Python, tutorialspoint* as found here https://www.tutorialspoint.com/machine_learning_with_python/index.htm or here www.tutorialspoint.com/.../machine_learning_with_python_tutorial.pdf

You'll have seen a number of instances by now in these notes where external sources are liberally referenced. Feel free to do this - make sure that you **always acknowledge your sources**.

We are going to work with the entire cleaned-up penguins data set that we originally stored in `dfp2`.

Let's remember what it looked like...

```
[40]: dfp2.head()
```

```
[40]:  species      island  bill_length_mm  bill_depth_mm  flipper_length_mm  \
0  Adelie  Torgersen         39.1           18.7           181.0
1  Adelie  Torgersen         39.5           17.4           186.0
2  Adelie  Torgersen         40.3           18.0           195.0
3  Adelie  Torgersen         36.7           19.3           193.0
4  Adelie  Torgersen         39.3           20.6           190.0

      body_mass_g      sex
0         3750.0    Male
1         3800.0  Female
2         3250.0  Female
3         3450.0  Female
4         3650.0    Male
```

We want to use the numerical features (values) in each row to predict species.

Before we start using the `sklearn` python library we need to see how we can pick these data items out using **array slicing**.

First, we can pick out the value of the species with this command (the colon part is important - it refers to column zero)

```
dfp2.iloc[2, 0:1].values
```

This refers to the entry in the third, the '2', row and first, the '0:1', column.

To refer to all rows we replace the 2 with a colon : - as we'll see below.

Let's see it in action...

```
[41]: dfp2.head()
```

```
[41]:  species      island  bill_length_mm  bill_depth_mm  flipper_length_mm  \
0  Adelie  Torgersen         39.1           18.7           181.0
1  Adelie  Torgersen         39.5           17.4           186.0
2  Adelie  Torgersen         40.3           18.0           195.0
3  Adelie  Torgersen         36.7           19.3           193.0
4  Adelie  Torgersen         39.3           20.6           190.0

    body_mass_g      sex
0      3750.0    Male
1      3800.0  Female
2      3250.0  Female
3      3450.0  Female
4      3650.0    Male
```

```
[42]: dfp2.iloc[2, 0:1].values
```

```
[42]: array(['Adelie'], dtype=object)
```

Second, we can refer to the four numerical features with this command

```
dfp2.iloc[1, 2:6].values
```

which refers to second row, and columns three to six inclusive. Once again we will use a colon to refer to all rows.

Again, let's see this in action...

```
[43]: dfp2.iloc[1, 2:6].values
```

```
[43]: array([np.float64(39.5), np.float64(17.4), np.float64(186.0),
          np.float64(3800.0)], dtype=object)
```

4.3 Using sklearn

We will now fit the k -NN model using the Manhattan, or taxicab, norm, which we also call the $p = 1$ norm:

$$\|x^* - x_i\|_1.$$

In addition, we will use two ($k = 2$) nearest neighbours, and we will also obtain something called the confusion matrix, and will print some performance data which allows us to assess the performance of our model.

Typically we assign the data set features to a variable called **X**, and the data set labels to a variable called **y**. Using the array slicing that we saw above this is straightforward...

```
[44]: # We assign the numerical features to X
X = dfp2.iloc[:, 2:6].values
# And we assign the species label to y
y = dfp2.iloc[:, 0].values
```

We could bifurcate the data into a training and test set ourselves, but **sklearn** provides a helper function for this. It is called **train_test_split**.

First we import it. Then we give it **X** and **y** and specify the proportion of the data that we use for the *hold out*, or *test* set. We also specify the *random state* for reproducibility, and *stratify* the selection so the same proportion of class labels as the input dataset appear in the train and test sets.

We'll specify that 40% of the data should be reserved for testing.

```
[45]: # from the scikit-learn library we use 40% of the data to test
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.40,
    random_state=42, stratify=y)
```

The function returns four subsets of data:

```
X_train - 60% of the data set features to be used to configure the model
X_test  - 40% of the data set features to used to test the configured model
y_train - 60% of the data set features matching the X_train features
y_test  - 40% of the data set features matching the X_test features
```

We can look at the sizes of each of these by using **shape** as follows...

```
[46]: print('shape of X_train = ', X_train.shape, ' and of X_test = ', X_test.shape)
print('shape of y_train = ', y_train.shape, ' and of y_test = ', y_test.shape)
```

```
shape of X_train = (199, 4) and of X_test = (134, 4)
shape of y_train = (199,) and of y_test = (134,)
```

Normalization of Data The next step is to normalize the feature data - the importance and role of this step is discussed in the recommended reading of pages 19 - 25 [MLF]. Again, **sklearn** provides a helper function for this called **StandardScaler**. This will remove the mean for each feature and scale it to unit variance. You can read more about this here: <https://scikit-learn.org/stable/modules/generated/sklearn.preprocessing.StandardScaler.html>

```
[47]: # import the helper and give it a name
from sklearn.preprocessing import StandardScaler
scaler = StandardScaler()
# initialise the scaler by feeding it the training data
scaler.fit(X_train)
# now carry out the transformation of all of the feature data
X_train = scaler.transform(X_train)
X_test = scaler.transform(X_test)
```

REMARK: note that `X_train` is used to provide the scaling data, and not `X_test`. This is because `X_test` is *hold out data*. We must treat it as **unseen**. We can freely transform it though, because that can be done without actually looking at it.

Fitting: Learning from Data We can now bring in the k -NN classifier method from `sklearn` and obtain a classifier object that uses $k = 2$ nearest neighbours and the $p = 1$ vector norm.

```
[48]: # import the k-NN classifier
from sklearn.neighbors import KNeighborsClassifier
# assign it with k=2 and p=1
classifier = KNeighborsClassifier(n_neighbors=2, p=1)
# give the training data to the classifier
classifier.fit(X_train, y_train)
```

```
[48]: KNeighborsClassifier(n_neighbors=2, p=1)
```

The last step above is just like the coloured cluster plots above **before** we plotted the larger crosses. The model now has *knowledge* of these clusters, this is an example of *machine learning*.

By giving the model the unseen test data we are in effect telling it where the large crosses are. The model then finds the two nearest neighbours, using the Manhattan norm, to classify the species of those crosses. This produces predictions of the species in `y_test`, and we call these predicted species values `y_pred`.

So, with the crosses as the features in the test set, we feed this in to the classifier and obtain the predicted values as follows...

```
[49]: y_pred = classifier.predict(X_test)
```

Evaluation of Performance Now we come to the real crux of the matter. We know what `X_test` should produce as species values - they are in `y_test`. What we actually get though are `y_pred`. If `y_pred = y_test` then we should be very happy because it indicates that the model works very well on unseen data.

In practice though, it is unlikely that each of the 134 elements in `y_pred` will match every one of the corresponding values in `y_test`.

Let's assess the quality of the model...

First we import the helper functions. Then we obtain and print the **confusion matrix**, next some statistics in a **classification report**, and then an **accuracy score**.

```
[50]: from sklearn.metrics import classification_report, \
        confusion_matrix, accuracy_score
cm = confusion_matrix(y_test, y_pred)
```

```
[51]: accsc = accuracy_score(y_test, y_pred)
print(f"Accuracy: {accsc}; Confusion Matrix:\n", cm)
clsrep = classification_report(y_test, y_pred)
print("Classification Report:\n", clsrep)
```

Accuracy: 0.9850746268656716; Confusion Matrix:

```
[[59  0  0]
 [ 2 25  0]
 [ 0  0 48]]
```

Classification Report:

	precision	recall	f1-score	support
Adelie	0.97	1.00	0.98	59
Chinstrap	1.00	0.93	0.96	27
Gentoo	1.00	1.00	1.00	48
accuracy			0.99	134
macro avg	0.99	0.98	0.98	134
weighted avg	0.99	0.99	0.98	134

We'll leave the classification report aside for now, and just note that the accuracy score tells us the proportion of the test set for which the species was correctly predicted.

What we want to spend some time on here is the confusion matrix.

The Confusion Matrix

```
[52]: print(cm)
```

```
[[59  0  0]
 [ 2 25  0]
 [ 0  0 48]]
```

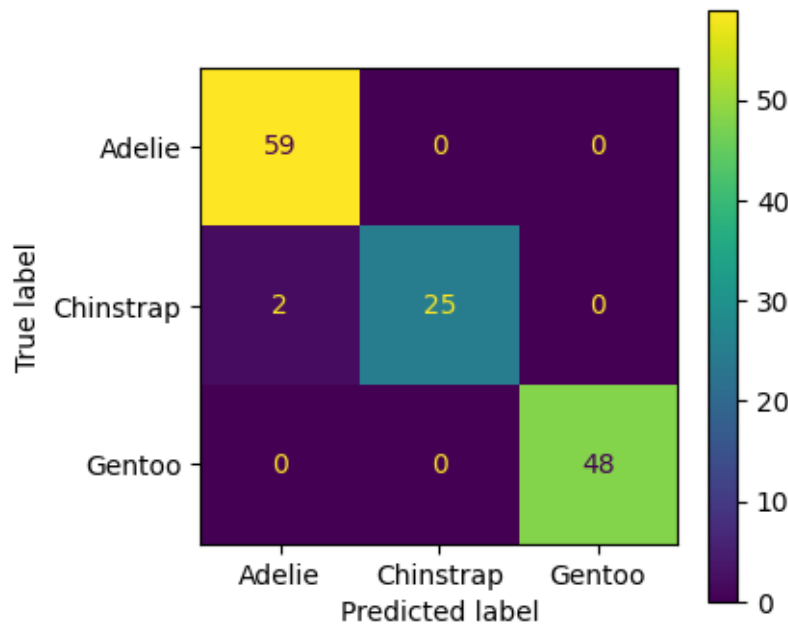
The confusion matrix is square with the same number of rows/columns as there are values for the label. In our case there are three possible label values: *Adelie*, *Chinstrap*, and *Gentoo*. We can refer to these as group 1, 2 and 3.

The entry in row i and column j of the confusion matrix tells us how many data points in X_{test} that were in group i were predicted by the model to be in group j .

Now, the representation of the confusion matrix above is a numpy array and although it is useful for coding, it isn't very user friendly. The following code gives us something much nicer, and it is much easier to understand.

```
[53]: from sklearn.metrics import ConfusionMatrixDisplay
cmplot = ConfusionMatrixDisplay(cm, display_labels=classifier.classes_)
```

```
[54]: cmplot.plot();
```



We get an immediate feel for *how good* the model is. The diagonal elements tell us how many species predictions match the true value. The off-diagonals tell us about the misses.

For example, the number in the middle of the top row tells us how many Adelie penguins were mistakenly predicted to be Chinstraps.

The accuracy is A/B for B the total of all entries and A the diagonal total. Compare the *Accuracy* score above.

4.4 Next Steps

We have now spent time getting familiar with key terms and concepts in Machine Learning

And we have seen how to configure, implement and assess an important classification algorithm: k -NN.

Our next task is to look briefly at *Logistic Regression*, a **binary classifier**.

We finish here with a few exercises you can use to consolidate your understanding.

Exercise Experiment with the k -NN classifier we just developed. For example,

- Change the 60%/40% bifurcation
- Change the value of k : decrease it to 1, or increase it to 3, 4, 5, ...
- Change the norm from $p = 1$ to $p > 1$.
- Does $p < 1$ make any sense here?

Exercise Use the following to generate some scatter plots

```
sns.scatterplot(data=dfp2, x="bill_length_mm", y="bill_depth_mm",
                style="species", hue="sex");

sns.scatterplot(data=dfp2, x="flipper_length_mm", y="body_mass_g",
                style="species", hue="sex");

sns.scatterplot(data=dfp2, x="body_mass_g", y="bill_depth_mm",
                style="species", hue="sex");

sns.scatterplot(data=dfp2, x="body_mass_g", y="bill_length_mm",
                style="species", hue="sex");

sns.scatterplot(data=dfp2, x="bill_length_mm", y="flipper_length_mm",
                style="species", hue="sex");

sns.scatterplot(data=dfp2, x="bill_depth_mm", y="flipper_length_mm",
                style="species", hue="sex");
```

Suppose we wished to predict gender from two features.

- What two features would work best do you think?
- Which pairs of features are unlikely to work well?

Exercise The confusion matrix we generated above is a **numpy array**. We will be looking in much more detail at these objects - both mathematically and in code - soon, but first here is a warm up. Let's recall the matrix:

```
[55]: cm
```

```
[55]: array([[59,  0,  0],
            [ 2, 25,  0],
            [ 0,  0, 48]])
```

We can use `cm[0,0]` to access the value in the first row and first column.

- What do you think `cm[1,1]` and `cm[2,2]` refer to?
- what do you think `cm[0,0]+cm[1,1]+cm[2,2]` produces?

Check your answers by using

- `print(cm[1,1],cm[2,2])`
- `print(cm[0,0]+cm[1,1]+cm[2,2])`

```
[56]: print(cm[1,1],cm[2,2])
      print(cm[0,0]+cm[1,1]+cm[2,2])
```

25 48

132

What do you think `cm.sum()` produces? Check, or discover, with

- `print(cm.sum())`

```
[57]: print(cm.sum())
```

134

How do you think `cm[0,0]+cm[1,1]+cm[2,2]` and `cm.sum()` relate to the Accuracy score given above? Print out your answer and check.

```
[58]: print((cm[0,0]+cm[1,1]+cm[2,2])/cm.sum())
```

0.9850746268656716

Compare `np.trace(cm)` to `cm[0,0]+cm[1,1]+cm[2,2]` - use your findings to shorten the command above

```
[59]: print(np.trace(cm)/cm.sum())
```

0.9850746268656716

4.5 Technical Notes, Production and Archiving

Ignore the material below. What follows is not relevant to the material being taught.

Production Workflow

- Finalise the notebook material above
- Clear and fresh run of entire notebook
- Create html slide show:
 - `jupyter nbconvert --to slides 02_knn.ipynb`
- Set `OUTPUTTING=1` below
- Comment out the display of web-sourced diagrams
- Clear and fresh run of entire notebook
- Comment back in the display of web-sourced diagrams
- Clear all cell output
- Set `OUTPUTTING=0` below
- Save
- `git add`, `commit` and `push` to FML
- copy PDF, HTML etc to web site
 - `git add`, `commit` and `push`
- rebuild binder

Some of this originated from

<https://stackoverflow.com/questions/38540326/save-html-of-a-jupyter-notebook-from-within-the-r>

These lines create a back up of the notebook. They can be ignored.

At some point this is better as a bash script outside of the notebook

```
[61]: %%bash
      NBROOTNAME='02_knn'
```

```
OUTPUTTING=1
```

```
if [ $OUTPUTTING -eq 1 ]; then
# jupyter nbconvert --to slides --theme=simple $NBROOTNAME.ipynb
jupyter nbconvert --to slides $NBROOTNAME.ipynb
cp $NBROOTNAME.slides.html ../backups/$(date +"%m_%d_%Y-%H%M%S")_$NBROOTNAME.
↪slides.html
mv -f $NBROOTNAME.slides.html ../formats/slides/

jupyter nbconvert --to pdf $NBROOTNAME.ipynb
cp $NBROOTNAME.pdf ../backups/$(date +"%m_%d_%Y-%H%M%S")_$NBROOTNAME.pdf
mv -f $NBROOTNAME.pdf ../formats/pdf/

jupyter nbconvert --to script $NBROOTNAME.ipynb
cp $NBROOTNAME.py ../backups/$(date +"%m_%d_%Y-%H%M%S")_$NBROOTNAME.py
mv -f $NBROOTNAME.py ../formats/py/
echo; echo 'Finished generating html, pdf and py output versions'
else
echo 'Not Generating html, pdf and py output versions'
fi
```

```
[NbConvertApp] Converting notebook 02_knn.ipynb to slides
[NbConvertApp] WARNING | Alternative text is missing on 7 image(s).
[NbConvertApp] Writing 406589 bytes to 02_knn.slides.html
[NbConvertApp] Converting notebook 02_knn.ipynb to pdf
/Users/simon/offline/research/ALCAS/Nilocas_repos/NilocasUV/lib/python3.14/site-
packages/nbconvert/utils/pandoc.py:50: RuntimeWarning: You are using an
unsupported version of pandoc (2.7.2).
Your version must be at least (2.9.2) but less than (4.0.0).
Refer to https://pandoc.org/installing.html.
Continuing with doubts...
    check_pandoc_version()
[NbConvertApp] Writing 92353 bytes to notebook.tex
[NbConvertApp] Building PDF
[NbConvertApp] Running xelatex 3 times: ['xelatex', 'notebook.tex', '-quiet']
[NbConvertApp] Running bibtex 1 time: ['bibtex', 'notebook']
[NbConvertApp] WARNING | bibtex had problems, most likely because there were no
citations
[NbConvertApp] PDF successfully created
[NbConvertApp] Writing 119783 bytes to 02_knn.pdf
[NbConvertApp] Converting notebook 02_knn.ipynb to script
[NbConvertApp] Writing 38517 bytes to 02_knn.py
```

Finished generating html, pdf and py output versions